Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/31/2019
PI Name:	Narayanan, Anand Ph.D.		
Project Title:	Simulated Microgravity-Induced Systemic Inflammation and Its Impact on Circulatory Function and Structure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	02/28/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Delp, Michael Ph.D. (MENTOR: Florida State University)		
Grant/Contract No.:	80NSSC19K0426		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP Long-term exposure to the weightless environment causes astronauts to become sick and develop issues like constipation, skin rashes, and increased likelihood of developing a cold. This microgravity environment also induces a redistribution of bodily fluids towards the head. This fluid redistribution causes their heart and circulatory system (i.e., the arteries, veins, and lymphatic vessels that transport blood and lymph throughout the body) to work differently. No one has explored whether the increased likelihood of astronauts developing colds/being sick is linked to changes in how the cardiovascular system functions in spaceflight. This is a concern, as how the immune system (which is responsible for making sure your body stays healthy) functions is linked to how well components of the immune system can travel throughout the body through the circulation. Therefore, our general hypothesis is that an inability for components of the		

Task Description:	immune system to circulate normally causes the increased likelihood of astronauts becoming sick, as well as also playing a role with how the circulatory system adapts to a weightless environment due to the consequences of impaired immunity. We will test this hypothesis in the gold-standard rodent model for simulating microgravity effects on Earth, the hindlimb-unloading (HU) model. Using HU, we will test long-term simulated microgravity effects on the digestive and hindlimb systems of the rodents for their changes in immune factors and circulatory system responses. This will be done utilizing techniques that can measure a broad-spectrum of immune factors in the tissues from the digestive system and hindlimb, to identify the most relevant target(s) that could influence altered immune response and changes in circulatory function. Furthermore, we will test the function of the arteries, veins, and lymphatics in the digestive and hindlimb systems to compare responses between the two organ beds in response to long-term simulated microgravity. Finally, we have discovered a novel therapeutic, anti-inflammatory agent, irisin, that we have shown in other rodent models of impaired immunity to resolve and beneficially improve outcomes of digestive and hindlimb structure, function, and immune state. We will use irisin as a countermeasure in our simulated microgravity studies to treat the predicted immune-driven circulatory changes in the digestive and hindlimb systems, for our long-term goal of a treatment that can holistically treat astronauts and disease conditions on Earth. These studies investigate novel and unexplored mechanisms, biological responses, and treatments in relation to NASA Space Biology's goals to understand the basic mechanisms that animals use to adapt/acclimate to spaceflight, and to contribute basic knowledge of biological adaptation/acclimation to spaceflight that may accelerate solutions to biomedical problems affecting human space exploration as well as human health on Earth. We will discover new knowledge addressing topics of interest including immunology, musculoskeletal changes, digestive system changes, vascular and lymphatic drainage, and adaptation of life to space. Indeed, through support of these research aims, we will discover new knowledge and therapies for the benefit of human space exploration and biomedical problems on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We will discover new knowledge and therapies for the benefit of human space exploration and biomedical problems on Earth.
Task Progress:	Long-term exposure to the weightless environment causes astronauts to become sick and develop issues like constipation, skin rashes, and increased likelihood of developing a cold. This microgravity environment also induces a redistribution of bodily fluids towards the head. This fluid redistribution causes their heart and circulatory system to work differently. No one has explored whether the increased likelihood of astronauts developing colds/being sick is linked to changes in how the cardiovascular system functions in spaceflight. This is a concern, as how the immune system (which is responsible for making sure your body stays healthy) functions is linked to how well components of the immune system can travel throughout the body through the circulation. Therefore, our general hypothesis is that an inability for components of the immune system to circulate normally causes the increased likelihood of astronauts becoming sick, as well as also playing a role with how the circulatory system adapts to a weightless environment due to the consequences of impaired immunity. We will test this hypothesis with the gold-standard model for simulating microgravity effects on Earth, the hindlimb-unloading (HU) model. Using HU, we will test long-term simulated microgravity effects on the digestive and hindlimb systems for their changes in immune factors and circulatory system responses. The Post-Doctoral Fellowship project is tied with NASA Grant: 80NSSC19K1322: Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure (PI: Dr. Michael Delp). This project has been delayed and did not start until more recently (Fall 2019); it includes the hindlimb unloading component of this fellowship project. Thus, the experimental work tied to this Post-Doctoral Fellowship has resulted in being delayed, and limited progress has been made to date.
Bibliography Type:	Description: (Last Updated: 05/22/2024)
Abstracts for Journals and Proceedings	Park H, Narayanan SA, Linarte M, Marulendra T, Mattia A, Dennis-Diaz D, Porcaro K, Dennis J, Behnke B, Muller-Delp J, Delp M. "Effects of Aging and Exercise Training on Vasomotor Function of Bone Resistance Arteries and Bone Blood Flow." To be presented at Experimental Biology 2020, San Diego, CA, April 4-7, 2020. FASEB Journal. 2020 Apr;34(1 Suppl). , Apr-2020